Synopsis of Datasets

1. Digits Dataset (Common Dataset)

The Digits dataset is a standard dataset available in the Scikit-learn library and is commonly used for handwritten digit recognition tasks. It consists of grayscale images representing digits from 0 to 9. The dataset is used to demonstrate the application of Singular Value Decomposition (SVD) for dimensionality reduction and classification.

Dataset Overview

• Total samples: 1797 images
• Number of classes: 10 (digits 0–9)
• Image size: 8 × 8 pixels
• Total features: 64 pixel intensity values
• Feature type: Numerical grayscale values
• Classification task: Multi-class classification

Each image is originally an 8 × 8 pixel matrix representing grayscale intensity values. For machine learning models, the image is flattened into a 64-dimensional feature vector.

Dimensionality Perspective

• The dataset contains 64 numerical features per sample.
• Each feature corresponds to the intensity value of a pixel in the image.
• Although the dimensionality is moderate, some features may be redundant or contain less useful information.
• Visualization of the dataset directly in 64-dimensional space is not possible.

Using Singular Value Decomposition (SVD), the dataset can be reduced to a smaller number of principal components while preserving most of the variance in the data.

High Dimensionality Considerations

1. Feature Redundancy
   Neighboring pixels in images often contain correlated information, leading to redundant features.
2. Curse of Dimensionality (Moderate Level)
   As dimensionality increases, the complexity of the classification model also increases.
3. Computational Cost
   Higher dimensional feature spaces require more computation for model training.
4. Feature Scaling Requirement
   Standardization was applied before applying SVD and the classification model.

Experimental Results

Accuracy Comparison:

• Accuracy without SVD: 0.9629629629629629

• Accuracy with SVD: 0.9388888888888889

Dimensionality Reduction:

• Features before SVD: 64

• Features after SVD: 20

Variance Analysis:

• Explained variance retained: 0.7931376270240807

Interpretation

SVD successfully reduces the dimensionality of the dataset while preserving most of the important information in the data. By reducing the number of features, the classification model can be trained more efficiently while maintaining comparable or improved accuracy. The results demonstrate that dimensionality reduction helps mitigate redundancy and improves computational efficiency.

2. Fashion-MNIST Dataset (Referred Dataset)

The Fashion-MNIST dataset is a large-scale dataset developed by Zalando Research as a replacement for the original MNIST handwritten digit dataset. It contains grayscale images of various fashion products belonging to different clothing categories.

The dataset is widely used for benchmarking machine learning algorithms in image classification problems.

Dataset Overview

• Total samples: 70,000 images

• Training samples: 60,000 images

• Testing samples: 10,000 images

• Number of classes: 10 clothing categories

• Image size: 28 × 28 pixels

• Total features: 784 pixel values

• Feature type: Grayscale pixel intensities

Class Labels

1. T-shirt / Top
2. Trouser
3. Pullover
4. Dress
5. Coat
6. Sandal

7. Shirt
8. Sneaker
9. Bag
10. Ankle Boot

Dimensionality Perspective

• Each image contains 784 features (28 × 28 pixels).
• The number of features is significantly larger than in the digits dataset.
• High dimensional feature spaces increase computational complexity.
• Direct visualization in 784-dimensional space is not feasible.

Applying SVD reduces the dimensionality of the dataset while retaining the most informative components.

High Dimensionality Considerations

1. Large Feature Space
   The dataset contains 784 features per sample, which increases model complexity.
2. Feature Correlation
   Pixels in images are often correlated with neighboring pixels.
3. Curse of Dimensionality (High Level)
   High dimensional feature spaces can make classification tasks more challenging.
4. Computational Cost
   Training models on large feature sets requires higher computational resources.
5. Feature Scaling Requirement
   Standardization was applied before dimensionality reduction and classification.

Experimental Results

Accuracy Comparison:
• Accuracy without SVD: *0.8475238095238096*
• Accuracy with SVD: *0.8213809523809524*

Dimensionality Reduction:
• Features before SVD: 784
• Features after SVD: 50

Variance Analysis:
• Explained variance retained: 0.8010697690788376

Interpretation

The Fashion-MNIST dataset represents a higher dimensional classification problem compared to the digits dataset. Applying SVD reduces the number of features significantly while preserving important patterns in the data. This helps improve computational efficiency

and allows machine learning models to focus on the most informative components of the dataset.

Summary Comparison

Dataset	Number of Features	Dimensionality Level	Key Insight
Digits Dataset	64	Moderate	Demonstrates dimensionality reduction using SVD on a small image dataset
Fashion-MNIST	784	High	Shows how SVD reduces large feature spaces while preserving important variance